

Automating *in operando* Optical Spectroscopy to Probe the Proximity Effect in 2D Materials

SURF 2026 Talk

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Introduction

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**Carnegie
Mellon
University**



Outline

- 1 What is the proximity effect? How will we study it?
- 2 Measuring Superconductivity with LabVIEW
- 3 Optical and Electrical simultaneously with LabVIEW
- 4 Conclusion, Q&A

1 Background

Superconducting Proximity Effect

Electrical

Superconductivity

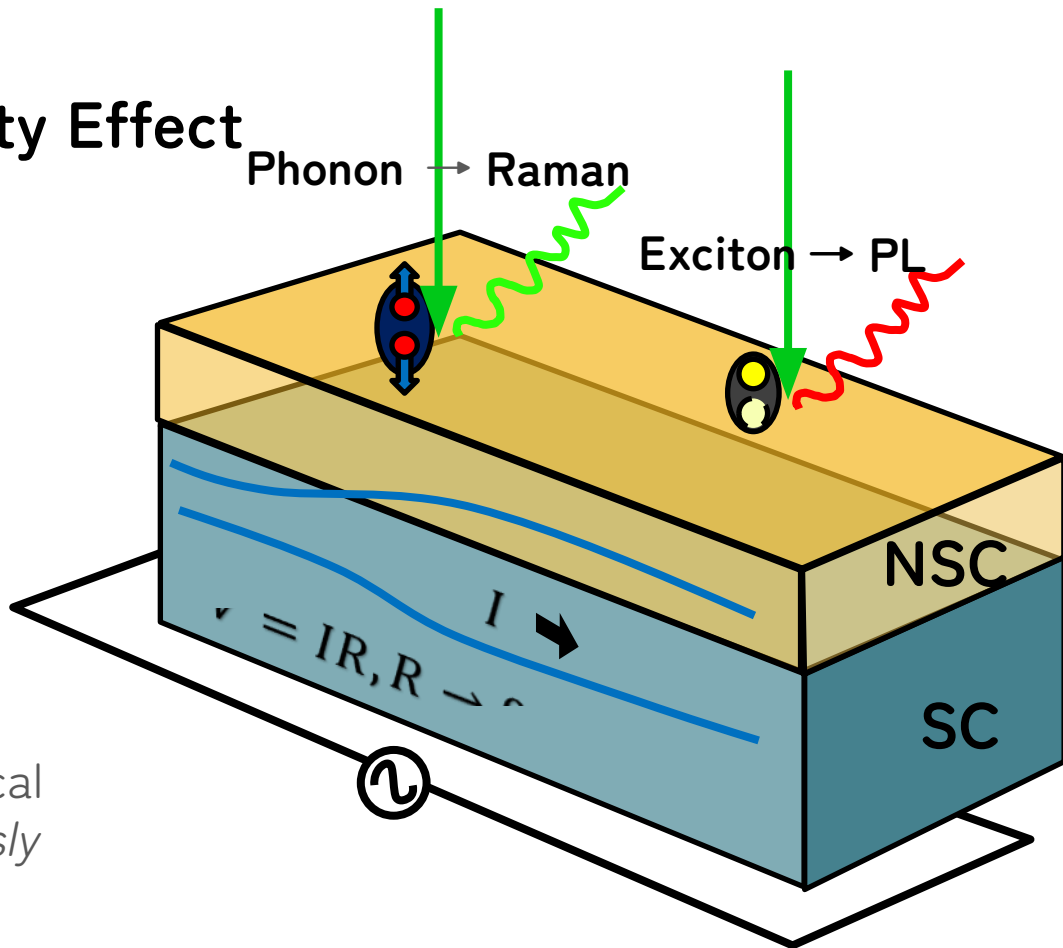
Proximity: Interface

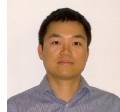
Optical

Phonons (vibration)

Excitons (electron/hole)

Big question: Can we observe optical changes in NSC while *simultaneously* measuring SC?





Son Le

Sample Materials

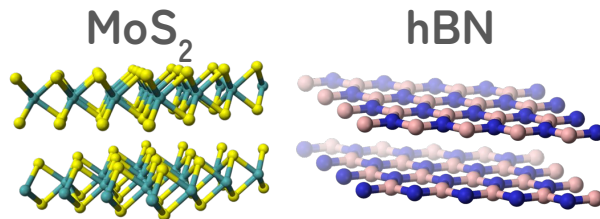
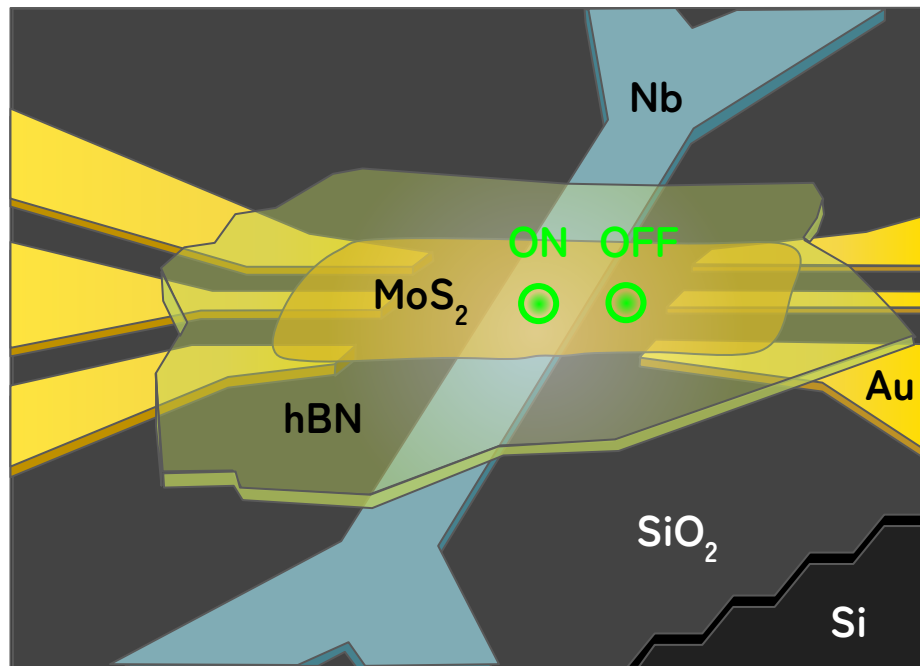
Area of interest: MoS₂ monolayer **ON** (proximity) and **OFF** (none) Nb strip.

Nb Superconductor (SC) ($T_c \sim 9$ K)

MoS₂ 2D semiconductor (NSC)

hBN “Blanket” layer to protect the MoS₂

SiO₂ Insulating substrate

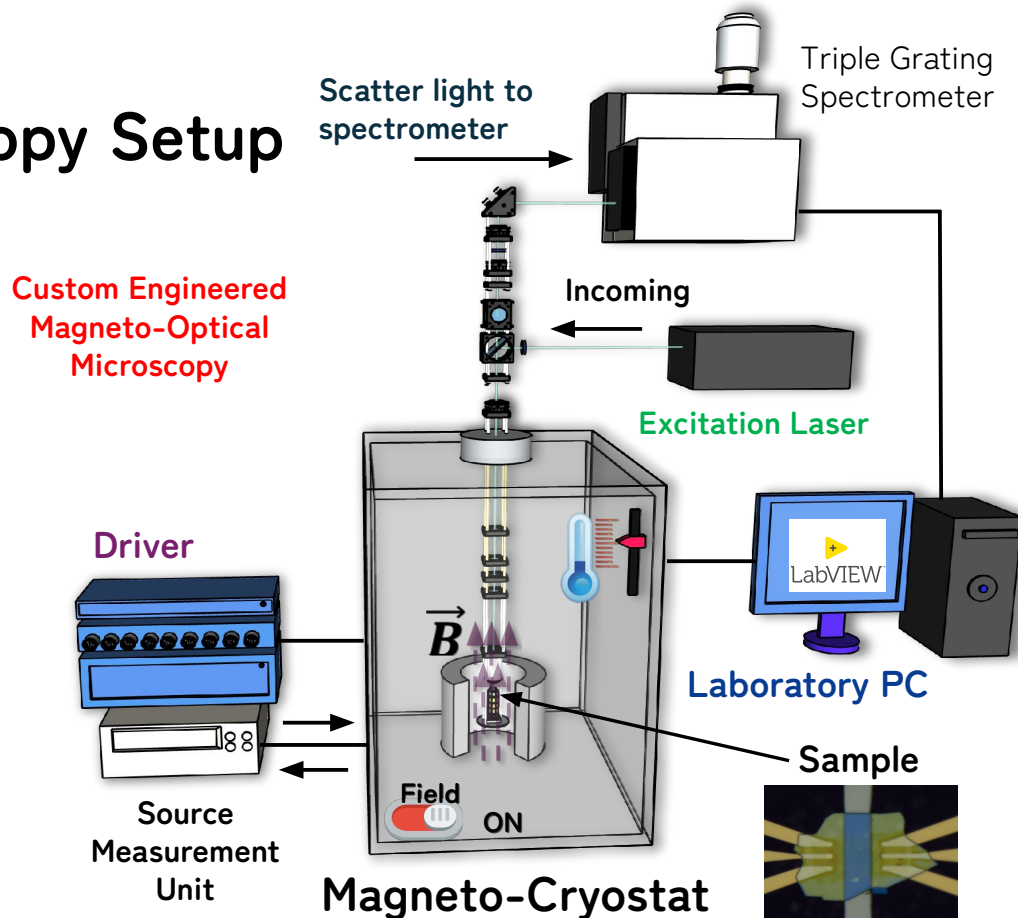


Overview of Spectroscopy Setup

Raman+PL spectroscopy w/
simultaneous...

- Magnetic field (0 to ± 9 T)
- Temperature control (~1.7 K to 300 K)
- Spatial mapping with X, Y, Z movement
 - Piezoelectric nanopositioners
- **Device transport (in operando)**

...automated on lab PC via LabVIEW



LabVIEW Overview

Virtual Instrument: block diagram & front panel → compiled to program

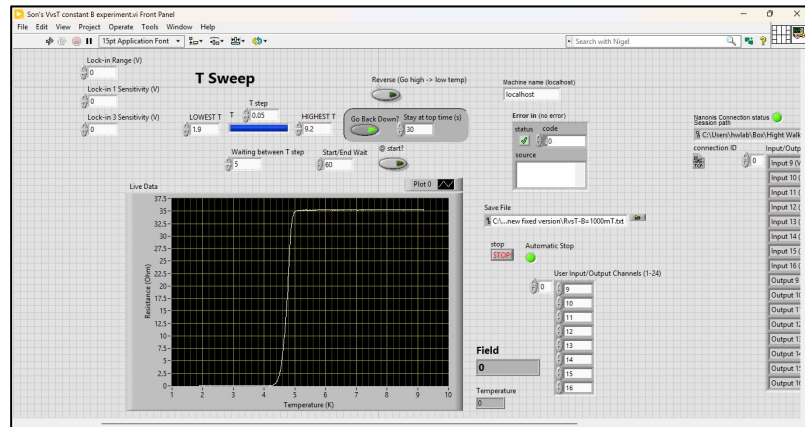
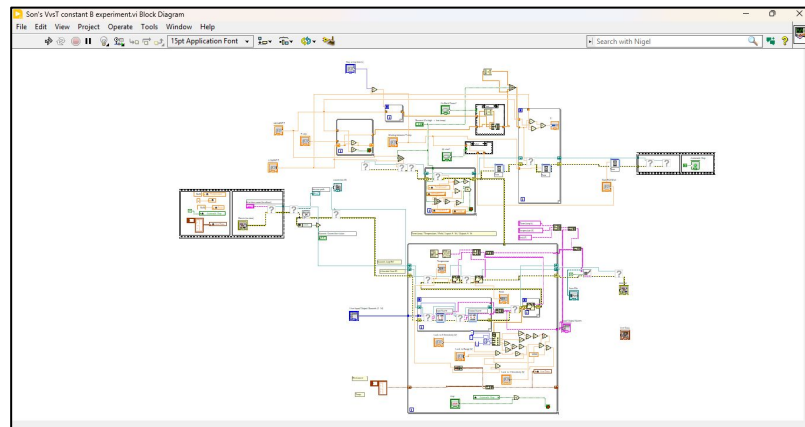
Block diagram: Flowchart connecting inputs, outputs, computations, actions.

Front panel: User interface.

Libraries for all different instruments



LabVIEW™
National Instruments



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Superconductor Performance

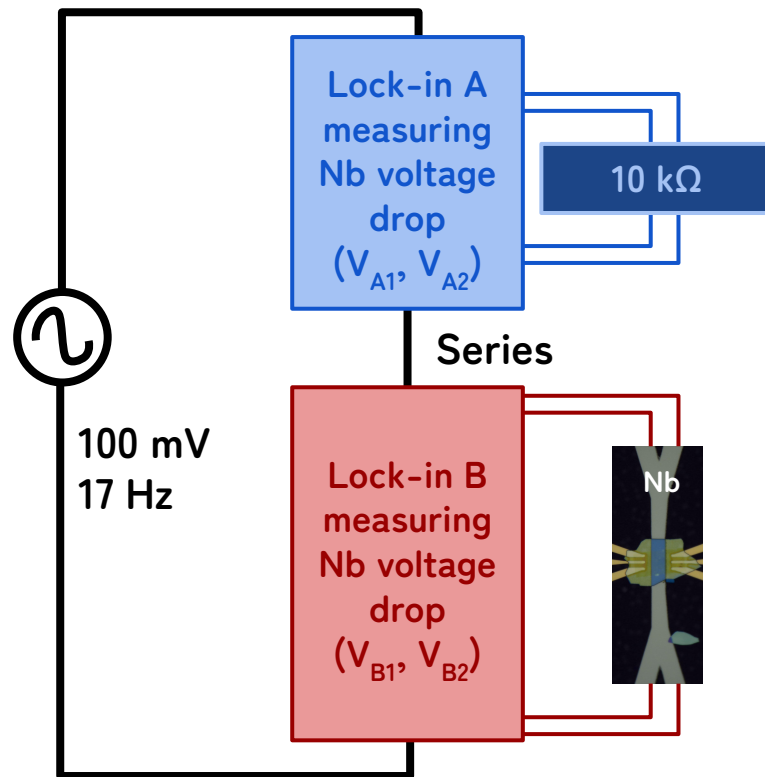
Electrical Measurement

- Lock-in Amplifier technique
- Four-probe measurement
- 2 steps:
Known $10\text{ k}\Omega \rightarrow \text{current} \rightarrow \text{resistance}$

$$I = \frac{V_R}{10\text{ k}\Omega}$$

$$V = IR$$

$$R_{Nb} = \frac{V_{Nb}}{I}$$



Methods - Resistance Measurement Automation

Voltage Collection

Collect 16 V values ~once/sec, also t , T , B .

Save to spreadsheet

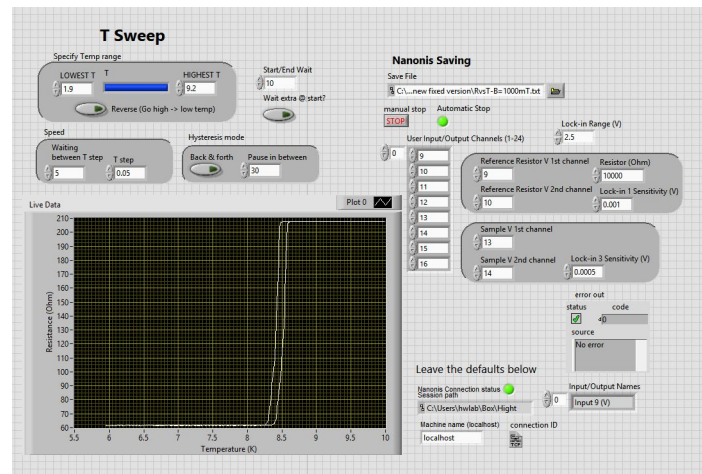
Temperature Ramping

Labview $\rightarrow$ Cryostat $\rightarrow$ Sample

Use small increments in T setpoint to ramp linearly

Field Ramping

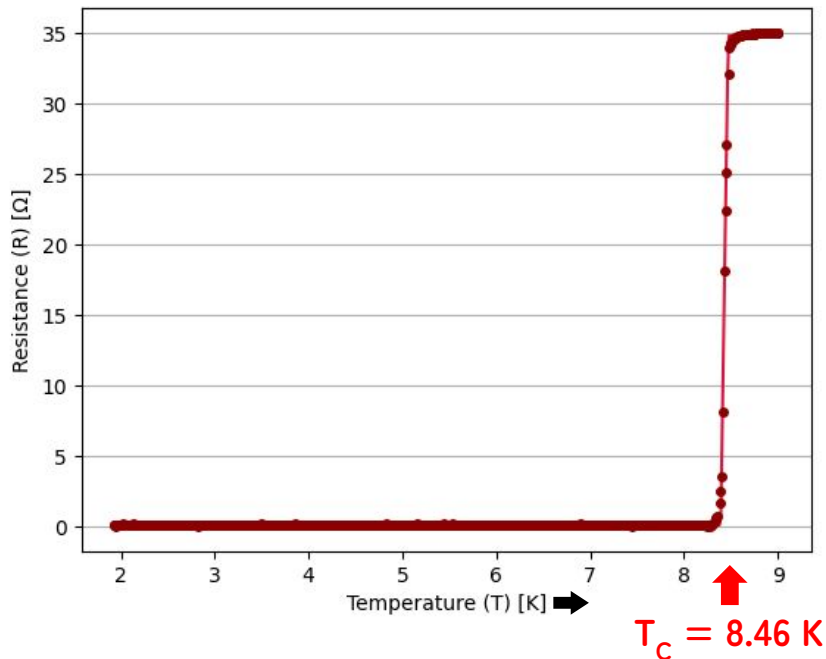
Set endpoints, magneto-cryostat automatically ramps.



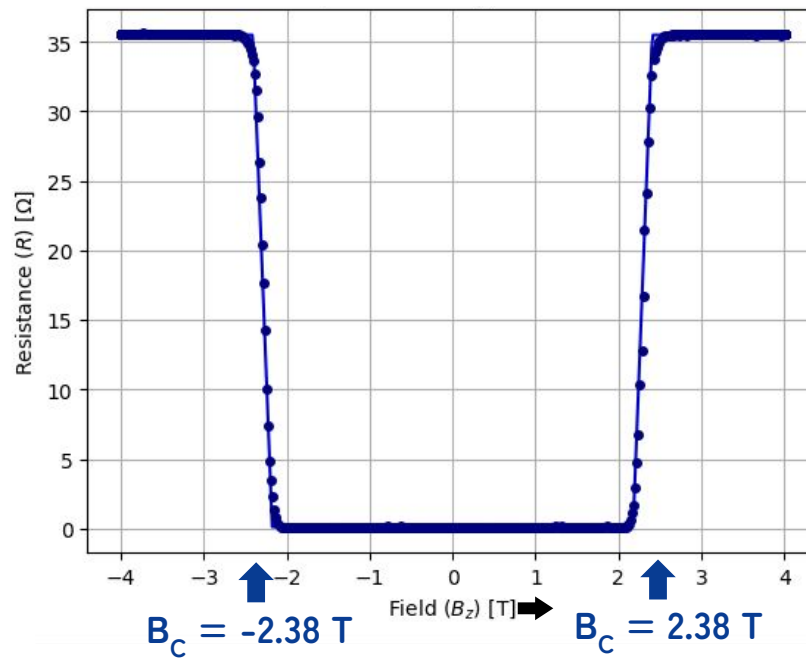
LabVIEW[™]
National Instruments

First time measuring superconductivity!

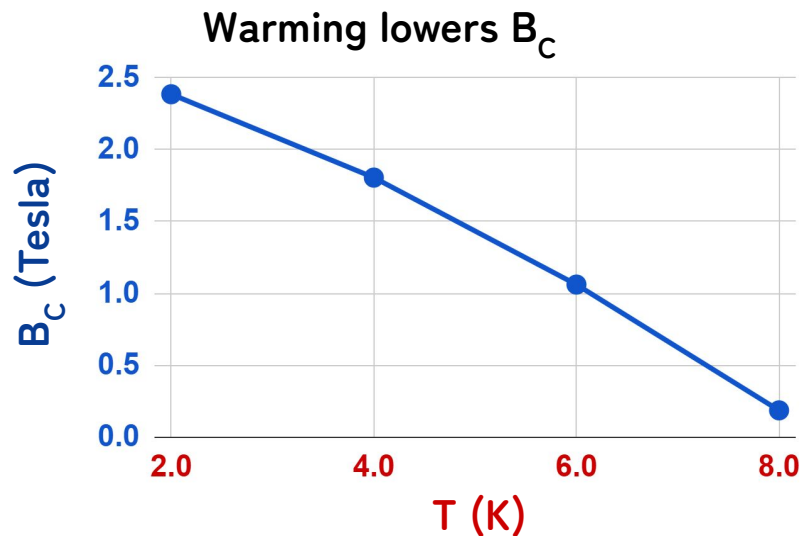
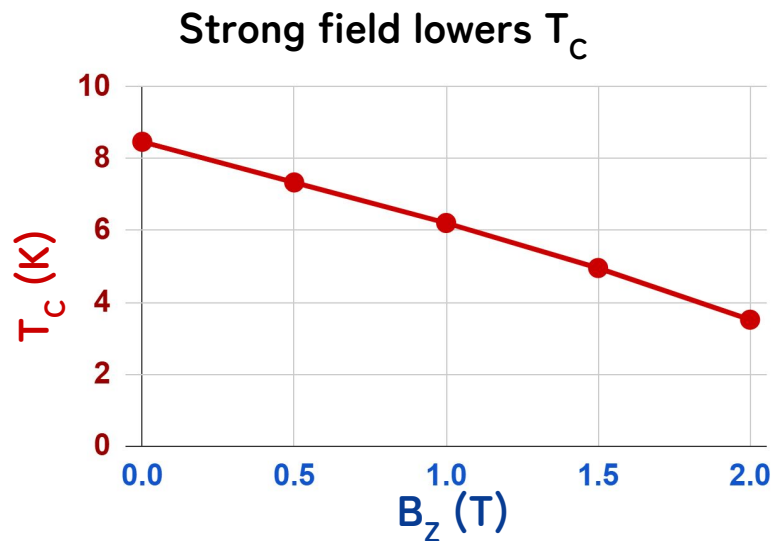
Superconducting Temperature ($B_z = 0$ T)



Superconducting Field ($T = 2$ K)



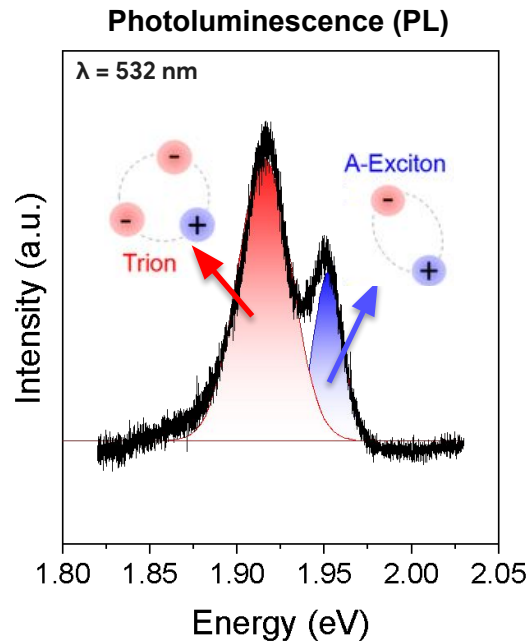
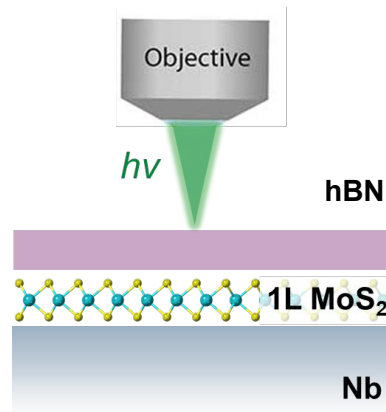
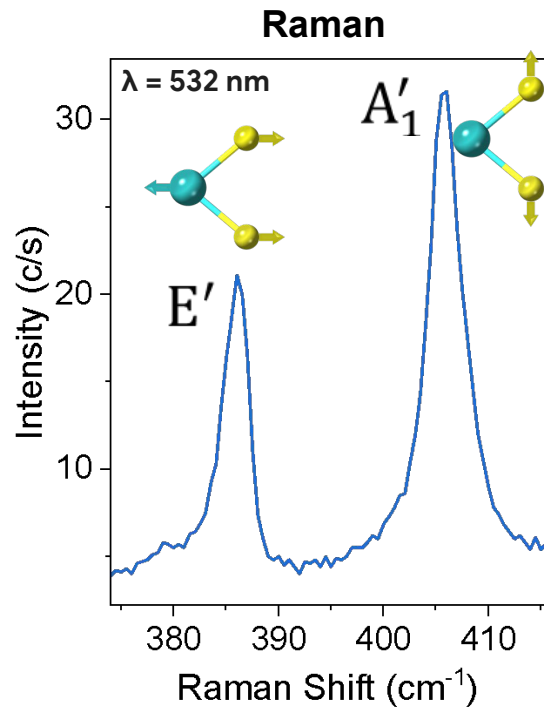
Electrical Results



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Optically Probing Proximity Effect

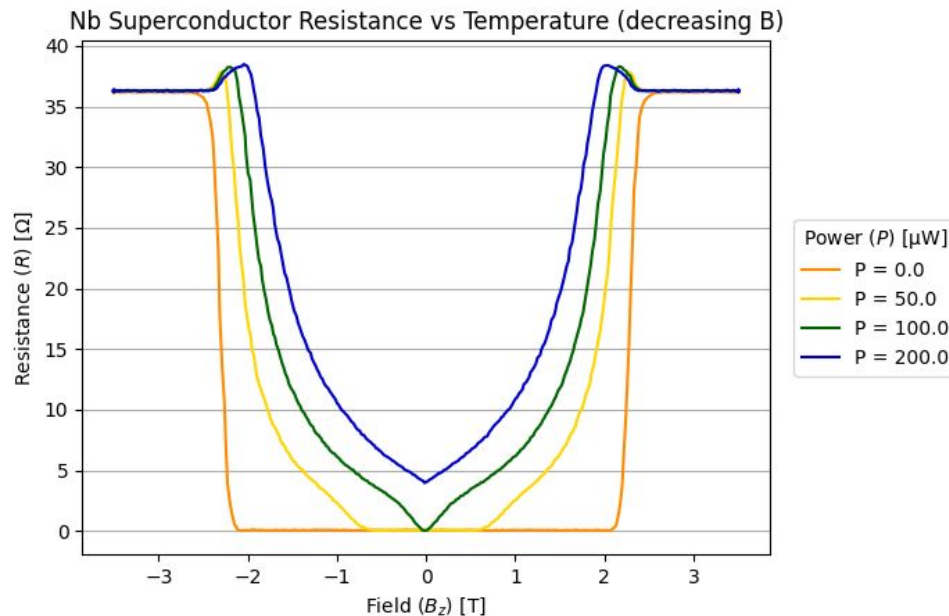
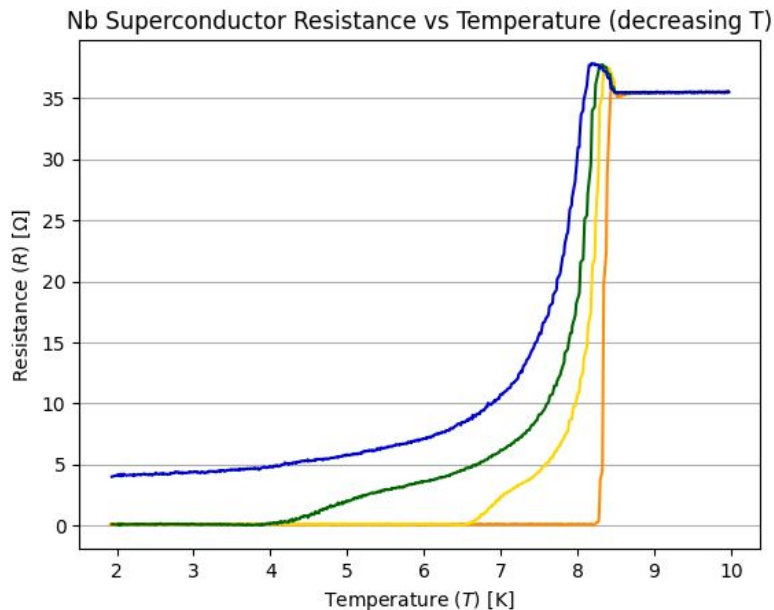
Optical Measurements



Thanks to Riccardo Torsi for graphics

Laser Power Heating

Even a weak laser pushes the sample back to non-superconducting!



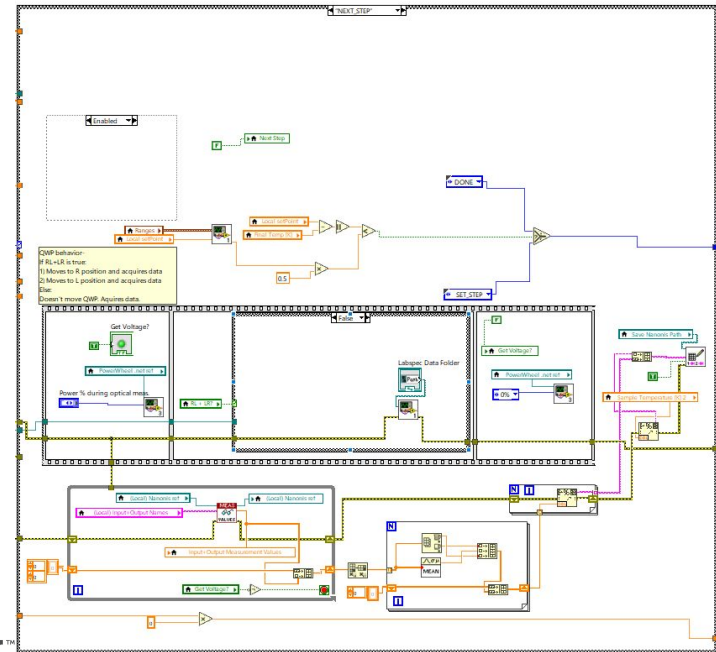
Simultaneous Electrical and Optical Measurements

- Existing program can take optical measurements for a range of temperatures
 - Waits for temperature to stabilize
- Modifications:
 - Measure voltages throughout optical measurement
 - Average voltages and save to spreadsheet
 - Block laser between optical measurements



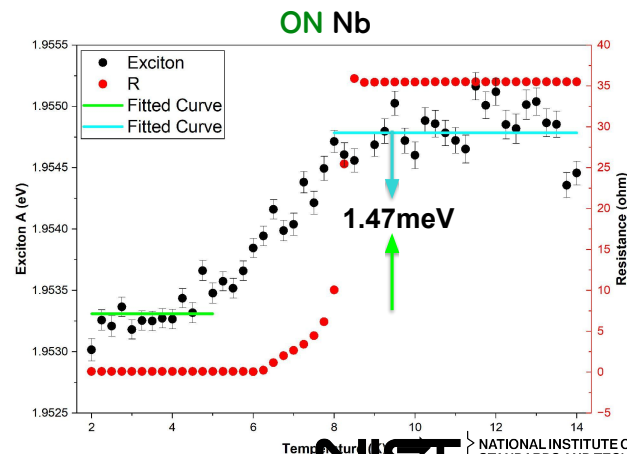
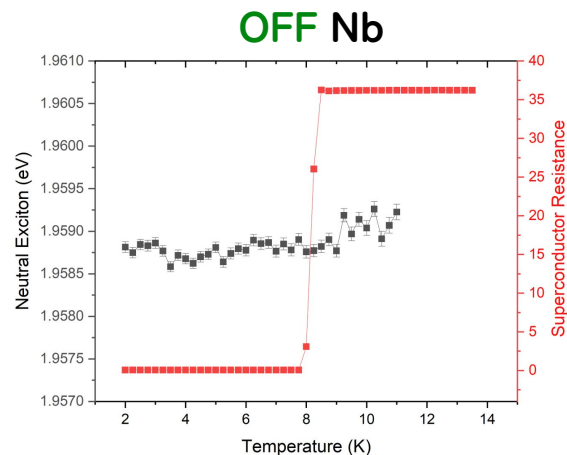
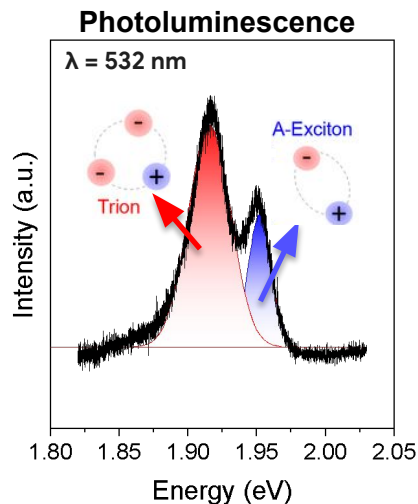
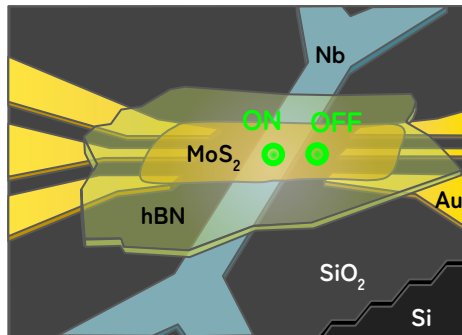
LabVIEW™

National Instruments



Simultaneous Optical/Electrical

- Big result: A-Exciton varies with proximity effect! **ON** vs **OFF** Nb?
- Diff = 1.47 meV
- Preliminary result



4

Conclusion, Q&A

Conclusions

Established correction of focus with Z-position via LabVIEW

Established electrical measurement capability via LabVIEW

Established simultaneous electrical and optical measurements via LabVIEW

Measured T_c with applied field in Nb

Measured optical evidence of proximity effect in Nb/MoS₂ (1.5 meV)



Acknowledgements

Thank you to the whole team!



References

Zarea M, Ueki H and Sauls JA (2023) Effects of anisotropy and disorder on the superconducting properties of niobium. *Front. Phys.* 11:1269872. doi: 10.3389/fphy.2023.1269872

Graphics

LabVIEW logo – Tanner Ochel

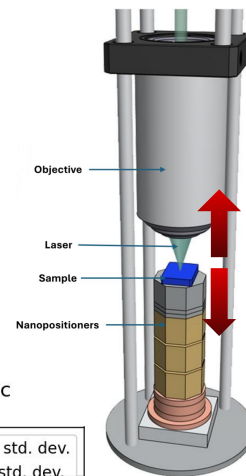
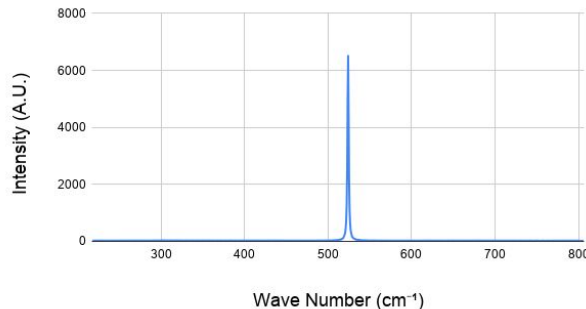
MoS₂ structure – Ben Mills

hBN structure – user benjah-bmm27, Wikipedia

Extra: Focus Correction

- Thermal effects $\rightarrow$ lost focus.
- Not very relevant for our Nb sample, other samples have different transitions
- New feedback loop: short measurements, piezo $\rightarrow$ greatest area (Gaussian peak fit)
- Smaller piezo voltage maintains small steps @ high T
- Good to 60K on clean Si
- Area for future improvements

Si Raman Spectrum



Intensity proportion change with uncorrected 532nm 2mW 5s 2acc and corrected 660nm 1mW 20s 4acc

